

ABSTRACT

This technical project begins by introducing the concept of aquaponics, including a brief history of its development and its place within the larger category of soilless culture and modern agriculture. It discusses the main theoretical concepts of aquaponics, including the nitrogen cycle and the nitrification process, the role of bacteria, and the concept of balancing an aquaponic unit. These project works on the three groups of living organisms (bacteria, plants and fish) that make up the aquaponic ecosystem. The most prevailing issues of the modern world are food and water crises. It is neither possible to consume the pesticide affected food nor grow one's own plants, due to scarcity of water and land. Under such conditions, there arises a need for a portable agricultural system which uses less water, space and is purely organic. One such solution is a small scale aquaponic system. This system is made by introducing an automation there by there is no need for setting aside extra time for system care. This project has used the data acquired from an existing aquaponic system to design and implement an effective small scale sustainable aquaponic system. This can lead to cost effective, sustainable ways of organic farming independent of the need for comparable land space requirement.

Dissolved chemical element is important for your fishes. Your lake establishes adequate levels of dissolved chemical element through correct aeration. Poor aeration is often one of the fastest and most silent killers of your fish. Though you've got had a long time lake for years, your fish would require a lot of chemical element as they grow larger. Night long your fish might die with fully no notice or symptoms. For your fish to thrive, it's counseled that your lake contains 7-9 mg/L of dissolved chemical element.

Water varies from place to position, reckoning on the condition of the supply water from that it's drawn and also the treatment it receives. The standard methodology of testing cloudiness, pH scale & Temperature is to gather samples manually and so send them to laboratory for analysis. However, it's been unable to fulfill the strain of water quality observance these days. Thus a collection of observance of cloudiness, pH scale & Temperature of Water quality has been developed. The system consists of cloudiness, pH scale & Temperature detector of water quality testing, observance accessories like sensors and controllers.

introduction

The project contains the methodology to build a aquaponic system suitable for different economic strata of the society especially focusing on the urban population where there is evident space and time constraints. This method contributes to one aspect of sustainable household development. In a small scale Aquaponic system, organic vegetables are cultivated in a limited space by recirculating water from a fish tank, rich in nutrients which are essential for the plant growth. Out of all the available water resources on planet Earth, 2.5% is freshwater resource. In this 2.5%, only 0.3% is the readily available freshwater resource accessible to humans. 70% of this limited amount of freshwater available is used for agriculture. Water scarcity already affects almost every continent and more than 40 percent of the people on our planet. By 2025, 1.8 billion people will be living in countries or regions with absolute water scarcity and two-thirds of the world's population would be living under water stressed conditions. In 2030, 47% of world population will be living in areas of high water stress. Most population growth will occur in developing countries, mainly in regions that are already experiencing water stress and in areas with limited access to safe drinking water and adequate sanitation facilities. Most of our food requires 100s of liters of water for production and adequate of per crop area for cultivation. The daily drinking water requirement per person is 2-4 liters, but it takes 2000 to 5000 liters of water to produce one person's daily food. In such a situation, a method like aquaponics which is the combination of hydroponics and aquaculture, can contribute effectively to the problem by lowering the amount of water usage for cultivation by 80% and also 75% of the area requirement. This project is mainly implemented on agriculture and production using natural as well as technical ideas. It is an innovative technique which is brought into play by combination of hydroponics and aquaculture. It can help to reduce oil usage. As the popularity in Aquaponics will increase, the need for chemical nutrients which are derived from oil will decrease, thus reducing oil usage. Aquaponics will reduce fresh drinking water consumption levels by being the most efficient use of water for growing vegetables. It has the most efficient use of nutrients out of all forms of agriculture.

BLOCK DIAGRAM

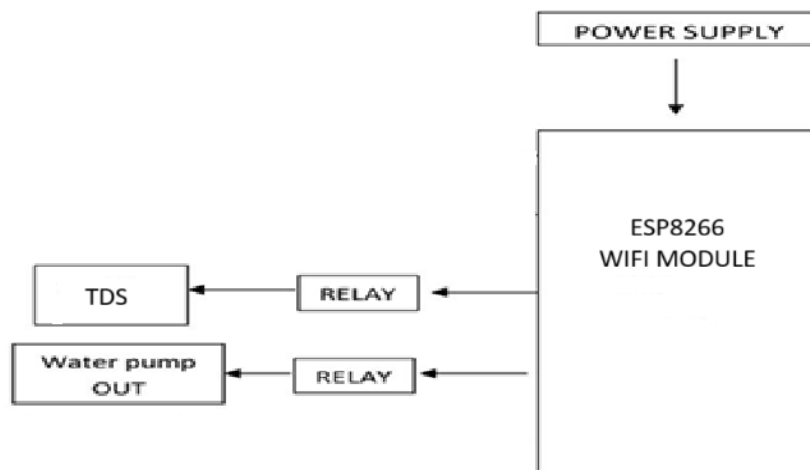


Figure 1:Block Diagram

WORKING:

In every aquaponics system, the first step is to establish a cycle in the system. So, for the formation of bacteria in the grow medium, the fish waste is circulated through the grow bed and from the grow bed the water is filtered and sent back to the tank. In a 7-day period the bacteria required to convert ammonia into the nitrate is formed. Then the plantation is started. Aquaponics is combination of hydroponics system and aquaculture system. In conventional hydroponics system plants required some mineral fertilizers. In conventional aquaculture system the water required periodical change, by combining these two systems into a bio-controlled looping formation the requirements of addition of mineral fertilizers and exchange of water is satisfied.

Large scale application of aquaponics system has not been reached a successful popularity and the concept is very complex so that people find difficulty to understand the concept of aquaponics like what is happening in aquaponics. Aquaponics can also applied in large scale system but it requires more investment to get an effective aquaponics system. This projects speaks of explaining the idea of aquaponics in terms of conventional methods of irrigations like canal irrigation. In this project the conventional canal irrigation system is modified for the large scale implementation of aquaponics system. Where all the limitations and applications are considered to implement in large scale system by modifying the

conventional irrigation system. For large scale irrigation process the canal irrigation in the form of conventional irrigation can be modified in the form of aquaponics agricultural irrigation system. Where the main canal serve as an aquaculture system and the irrigation fields serves as a hydroponics system. Here the main can has the ability to serve as aquaculture system by receiving large amount of water from canal head work. At the same time the nutrient rich water has been produced by fishes in main canal is discharged to irrigation filed. After the process of uptake by plants the discharged water received by collecting system. Then the water again recirculated into main canal with proper controlling of water quality parameters. In irrigation field racking system also applicable by using this method. Prototype of aquaponics system is made for the observation of behaviours of both plants and fishes.

ESP8266 WIFI MODULE



Figure 2: ESP8266

The ESP8266 is a low-cost Wi-Fi-enabled microcontroller used for IoT applications and wireless communication. It supports connecting to existing Wi-Fi networks or creating its own hotspot for data transfer. With built-in TCP/IP protocols and programmable GPIO pins, it can interface with sensors and control devices easily. Its small size, low power consumption, and easy programming make it ideal for smart farming and automation projects.

PIN CONFIGURATION

NODE MCU	GPIO No	Type / function	Description
D0	GPIO16	Digital I/O	No PWM, no interrupts; used for wake-up
D1	GPIO5	Digital I/O, I2C SCL	Commonly used for sensors/motors
D2	GPIO4	Digital I/O, I2C SDA	Used for sensors & modules
D3	GPIO0	Digital I/O	Must be high at boot; avoid using for output
D4	GPIO2	Digital I/O, Built-in LED	Must be high at boot; avoid low at startup
D5	GPIO14	Digital I/O, SPI SCK	Safe for sensors and modules
D6	GPIO12	Digital I/O, SPI MISO	Safe for general use
D7	GPIO13	Digital I/O, SPI MOSI	Safe for general use
D8	GPIO15	Digital I/O, SPI CS	Must be LOW at boot; avoid using for inputs
A0	ADC0	Analog Input	Reads 0–1V input; used for TDS sensor
Vin	-	Power Input	5V supply input
3V3	-	Power Output	3.3V for sensors
GND	-	Groun0064	Common ground
RST	-	Reset Pin	Resets the module
EN	-	Chip Enable	Must be HIGH for normal operation

Table 1 : ESP8266 PIN CONFIGURATION

FEATURES

- Has built-in Wi-Fi.
- Uses low power.
- Connects easily to sensors.
- Easy to program with Arduino.

TDS SENSOR

A TDS sensor measures the Total Dissolved Solids in aquaponics water, indicating the levels of nutrients, salts, and minerals available to plants. In a healthy aquaponics system, TDS typically stays between 300–600 ppm, depending on the plant and fish species. Monitoring TDS ensures proper nutrient availability for plants while keeping the water safe for fish. Accurate readings require stable water temperature and converting EC values to TDS using a factor (commonly 0.5 or 0.7). Regular calibration with standard solutions is essential to maintain reliable sensor performance.



Figure 3: TDS SENSOR

Sensor Pin Description

SL NO	PINS	PIN DESCRIPTION
1	A0	Analog output
2	GND	Ground
3	VCC	Power supply(3.3v)

Table 2 : TDS Sensor Pin Description

FEATURES:

- TDS shows how much stuff (nutrients) is in the water.
- It helps check if the water is healthy for fish and plants.
- TDS is linked to water's ability to carry electricity.
- Right TDS keeps fish safe and plants growing well.

RELAY :

A relay in an aquaponic farming system is used to safely control the water pump by isolating the low-voltage control circuit from the high-voltage pump load. It receives an electrical signal from a microcontroller or timer and switches the pump on or off based on system needs. This ensures precise water circulation between the fish tank and grow bed. By automating flow control, the relay helps maintain stable nutrient distribution and oxygen levels. Overall, it increases system reliability while reducing manual intervention.



Figure 4: RELAY

Relay Pin Description

SL NO	PINS	PIN DESCRIPTION
1	D5	GPIO14
2	GND	Ground
3	VCC	Power supply(3.3v)

Table 3 :Relay Pin Description

FEATURES:

- Relays act as automatic switches in aquaponic systems.
- They control pumps using small currents.
- Protect equipment from overload or damage.
- Help run the system safely and efficiently without manual operation.
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WATER MOTOR:

The water pump is a key component in aquaponic systems, circulating water between the fish tank and grow beds. It ensures continuous flow, delivering nutrients from fish waste to the plants and maintaining oxygen levels for fish. Controlled by a relay and microcontroller, the pump can operate automatically based on system requirements. Proper pump selection is crucial, considering flow rate, head height, and power efficiency. Overall, it supports a stable and healthy environment for both plants and fish, enabling efficient and sustainable aquaponic farming.



Figure 5: WATER MOTOR

FEATURES:

- Pumps water to circulate between fish tanks and plant beds.
- Provides oxygen-rich water for fish and nutrients for plants.
- Helps maintain water flow and system balance.
- Runs automatically to keep the farm healthy and efficient.

POWER SUPPLY



Figure 6: 12VDC – 1.5A POWER ADAPTER

Specifications:

- Input Voltage : 100~240V
- Output Power : 12W
- Output Voltage : DC 12V $\pm 5\%$
- Output Current : DC 1.5A
- Frequency : 50/60 Hz
- AC Inlet Mode : 2Pins AC Cord

SOFTWARE DESIGN

HARDWARE RESOURCE ALLOCATION

The aquaponic system uses several hardware components that together monitor water quality, control feeding, and manage plant irrigation. The allocation of each hardware resource is as follows:

ESP8266 Node MCU (Main Microcontroller + Wi-Fi)

- Acts as the central controller for the entire aquaponic system.
- Performs:
 - Reading TDS values
 - Controlling relays for pumps and feed motor
 - Sending live sensor data to Blynk IoT
- Built-in Wi-Fi eliminates the need for an external IoT module.

Sensors

a. TDS Sensor (Total Dissolved Solids)

- Measures water quality inside the fish tank.
- Detects:
 - Nutrient concentration
 - Fish waste levels
 - Water purity for plant and fish health
- Sends digital/ analog values to ESP8266.

PROGRAMMING LANGUAGE

The programming language used for developing this aquaponic farming system is Embedded C / Arduino C. It is a simplified version of the C/C++ programming language, specially designed for microcontrollers such as the ESP8266, ESP32, and Arduino boards.

Key Features of This Programming Language in Code

1.The language supports Wi-Fi libraries for sending data to Blynk.

- Sensor and Motor Control

2.we can easily read analog values from the TDS sensor and control the water pump using digital pins.

- Real-time automation

3.The code continuously monitors TDS levels and automatically switches the pump ON/OFF.

DEVELOPMENT TOOLS

The development of the automatic aquaponic system requires both software and hardware tools to design, program, simulate, and test the IoT-based automation. The tools used in this project are:

Arduino IDE

- Primary development environment for writing and uploading code to the ESP8266 Node MCU.
- Supports C/C++ programming.
- Provides library management for TDS, temperature, moisture sensors, Wi-Fi, and Blynk integration.

ESP8266 Board Manager

- Installed within Arduino IDE.
- Allows compiling and uploading code specifically for:
 - Node MCU ESP8266
 - ESP-12E modules
- Required to configure Wi-Fi and Blynk IoT functions.

Blynk IoT Platform

- Used to build the mobile dashboard for:
 - Monitoring TDS values
 - Reading moisture and temperature
 - Controlling water pumps and feed motor
- Provides API commands, virtual pins, and cloud connectivity.

Libraries Used

Several embedded libraries support sensor reading and IoT functions:

- ESP8266WiFi.h – Connects microcontroller to Wi-Fi
- BlynkSimpleEsp8266.h – Cloud data transfer
- TDS Sensor library / Analog processing functions – For TDS measurement
- BlynkTimer – Used for periodic pump running and feeding cycles

Serial Monitor / Serial Plotter

- Tools within Arduino IDE used to:
 - Debug sensor readings
 - Monitor TDS, temperature, and moisture value

Design

The design of the automatic aquaponic system integrates hardware, software, and IoT components to create a fully automated plant and fish-growing ecosystem. The system is engineered to continuously monitor water quality, regulate nutrient circulation, and automate feeding cycles while providing real-time data through the internet. The design consists of the following major elements:

System Architecture Design

The system is designed using a recirculating loop that combines aquaculture and hydroponics:

- **Fish tank:**
Acts as the nutrient source, where fish waste increases TDS and enriches the water.
- **Grow Bed Tray:**
Plants absorb nutrients and purify the water.
- **Water Pumps (IN & OUT):**
Maintain circulation between fish tank and plant bed.
- **ESP8266 Node MCU:**
Serves as the main controller, gathering sensor data and automating all system functions.

Sensor Integration Design

Sensors are placed strategically to monitor critical parameters:

TDS sensor:

Installed in the fish tank to measure dissolved solids indicating water quality

Control System Design

The control system uses the ESP8266 microcontroller to automate mechanical operations:

- **Relay Module:**
Controls high-voltage Devices (pumps) from the low-voltage ESP8266.
- **Pump Operation Design:**
Pumps turn ON when:
 - Water circulation timer triggers
 - TDS exceeds safe limits

IoT Connectivity Design

The system includes full IoT integration:

- ESP8266 connects to Wi-Fi.
- Sends real-time data (TDS) to Blynk IoT.
- User can:
 - Monitor water quality anywhere in the world.
 - Turn pumps ON/OFF remotely.
 - Receive alerts for abnormal TDS.
- Ensures fully remote monitoring and control.

Power Supply Design

- ESP8266 powered using 5V USB or regulated adapter.
- Pumps and feed motor powered via 12V or 5V depending on type.
- Relays isolate microcontroller from high-power circuits for safety.

Software Design

The software is structured into modules:

- Sensor reading module
- TDS calculation algorithm
- Decision-making (automation logic)
- Relay control module
- Blynk IoT communication module
- pump timer

This modular design ensures easy debugging and future upgrades

RESULT

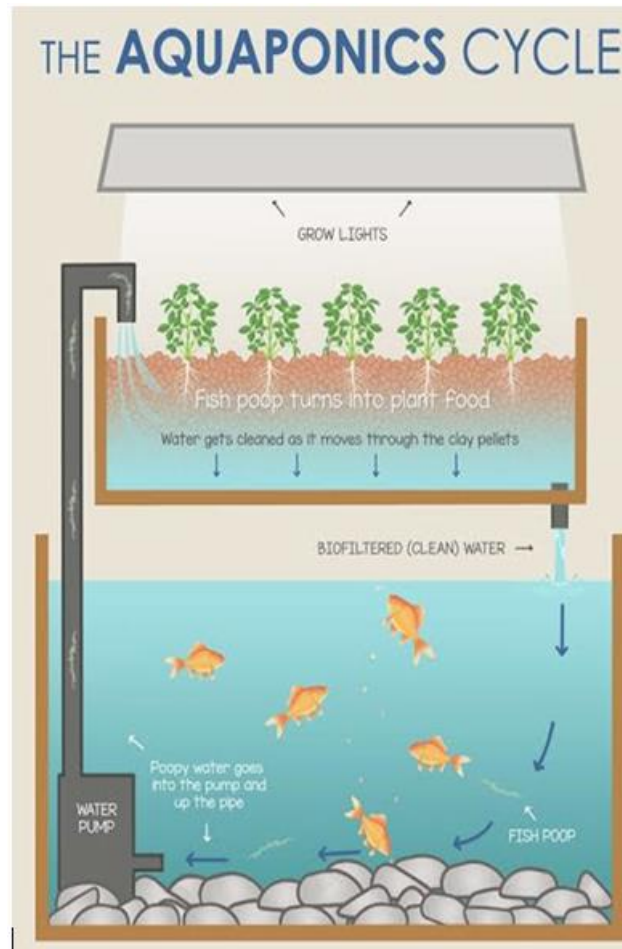


Figure 8:RESULT OF AQUAPONICS FARMING

The aquaponic farming system was successfully developed and tested using a TDS sensor, ESP8266 microcontroller, water motor, relay, and Blynk IoT platform. The system continuously monitored water quality and maintained suitable conditions for plant and fish growth.

The TDS sensor accurately measured the Total Dissolved Solids level in the water. Based on the sensor readings, the water motor operated automatically through the relay to maintain the required nutrient balance. All readings were successfully displayed on the Blynk application in real time. The system maintained stable TDS values within the recommended range for aquaponics, showing that the automation effectively controlled water quality. The plants showed healthy growth because the nutrient levels were automatically managed, and the fish remained healthy due to clean, balanced water .

CONCLUSION AND FUTURE SCOPE

CONCLUSION:

Effective aquaponic farming integrates aquaculture and hydroponics to create a self-sustaining ecosystem. Fish waste provides essential nutrients for plants, while plants naturally filter and purify the water. The TDS (Total Dissolved Solids) sensor monitors water quality, ensuring nutrient levels remain optimal for plant and fish health. Relays control the water pump automatically, maintaining consistent circulation between the fish tank and grow beds. This automation reduces manual labor and prevents system failures due to irregular water flow. Continuous monitoring and pump control promote healthy plant growth and higher crop yields. The system conserves water compared to traditional farming, making it environmentally sustainable. By balancing fish and plant requirements with automated sensors and controls, aquaponics maximizes productivity in limited space. Early detection of water quality issues ensures system reliability and prevents losses. Overall, the integration of sensors, relays, and pumps makes aquaponics an efficient, eco-friendly, and scalable method for sustainable food production.

FUTURE SCOPE:

OBJECTIVES & SCOPE OF STUDY

- 1) To design a simple structure
- 2) To reduce the human efforts
- 3) To give the human comfort
- 4) To make a device in which we can get the data anywhere in globe
- 5) To grow the plants without soil